Simulation-Based Inference for Plant-Wide Fault Diagnosis and Artefactual Non-Identifiability Under Closed-Loop Control

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Fault diagnosis in feedback-controlled chemical processes is fundamentally limited by the controller's tendency to compensate for degradation, suppressing the parametric signatures that estimation methods rely on. We apply simulation-based inference (SBI) across two systems of increasing complexity: a PI-controlled CSTR (two fault parameters) and the Luyben reactor-column-recycle benchmark (five fault parameters, three PI loops, liquid recycle). SBI approximates the Bayesian posterior over degradation parameters by a neural spline flow trained on summary statistics of fixed-length observation windows. Identifiability is characterised via Fisher information analysis, validated with simulation-based calibration, and benchmarked against extended Kalman filter and Markov chain Monte Carlo baselines. For the CSTR, the matched closed-loop posterior accurately recovers catalyst decay and jacket fouling in non-saturating scenarios, while local sensitivity analysis shows that PI temperature control makes jacket conductance less well conditioned than catalyst activity. Severe fouling introduces actuator-saturation undercoverage, and temperature-measurement bias creates a practical mild-fouling confound near the classification boundary. For the recycle plant, three apparent joint non-identifiabilities under summary- statistic compression are shown, using a raw-trajectory Fisher-information check, to be representation artefacts rather than plant-level limitations. In contrast, thermal masking by integral temperature control remains a genuine representation- independent information loss. We show that confounds of comparable magnitude can have very different operational consequences, depending on the fault taxonomy. Furthermore, we show that amortised SBI can perform closed-loop, multi-unit fault identification directly from simulator-generated training data, without historical labelled-fault records. SBI relies only on a simulator and is applicable when no tractable likelihood exists. It can represent joint, non-Gaussian posterior structure induced by closed-loop recycle dynamics, while amortising inference to real-time monitoring cadence.

Keywords: condition monitoring, Bayesian statistics, simulation-based inference, predictive maintenance, closed-loop identifiability, recycle process, prognostics and health management, fault diagnosis.

I. INTRODUCTION

Continuous chemical processes such as reactor-separator-recycle configurations are subject to progressive degradation: catalyst activity decays through poisoning or sintering, heat-exchanger surfaces accumulate fouling deposits, and separator efficiency erodes over operating cycles. Digital twins can provide a real-time, physics-informed digital representation of these processes, enabling predictive maintenance and proactive fault management [2, 3].

Timely quantification of such degradation is essential for risk-informed maintenance scheduling, product quality assurance, and avoidance of unplanned shutdowns [4, 5]. Current industrial practice relies on a few broad families of tools for the diagnosis of process faults [6, 7]: soft sensors, which regress fault parameters from process measurements but do not provide uncertainty quantification [8, 9], model-based observers, principally the extended and unscented Kalman filters (EKF/UKF), which propagate Gaussian approximations of the joint state-and-parameter posterior [10, 11], multivariate statistical methods based on principal component analysis or partial least squares, which provide sensitivity indices and alarms but not calibrated parameter posteriors [7, 12–14], and data-driven anomaly detectors built on neural or recurrent architectures, which flag deviations from normal operation without quantifying fault severity in physically interpretable units [15–19]. None of these approaches delivers a calibrated probability distribution over the fault-parameter space needed for maintenance planning. The decision to schedule an intervention depends on the probability that a parameter has crossed a critical threshold, not on whether a point estimate has done so.

This diagnostic task is made structurally harder by the feedback controllers that regulate the plant. Decentralised proportional-integral (PI) loops compensate for degradation. It is a well-understood limitation that the control compensation suppresses the same parametric signatures that estimation methods rely on [20, 21]. Critically, this method-independent limitation is rooted in the information content of the closed-loop data, and it affects Bayesian and frequentist, online and offline, neural and classical methods equally (see Ljung 22, Ch. 13).

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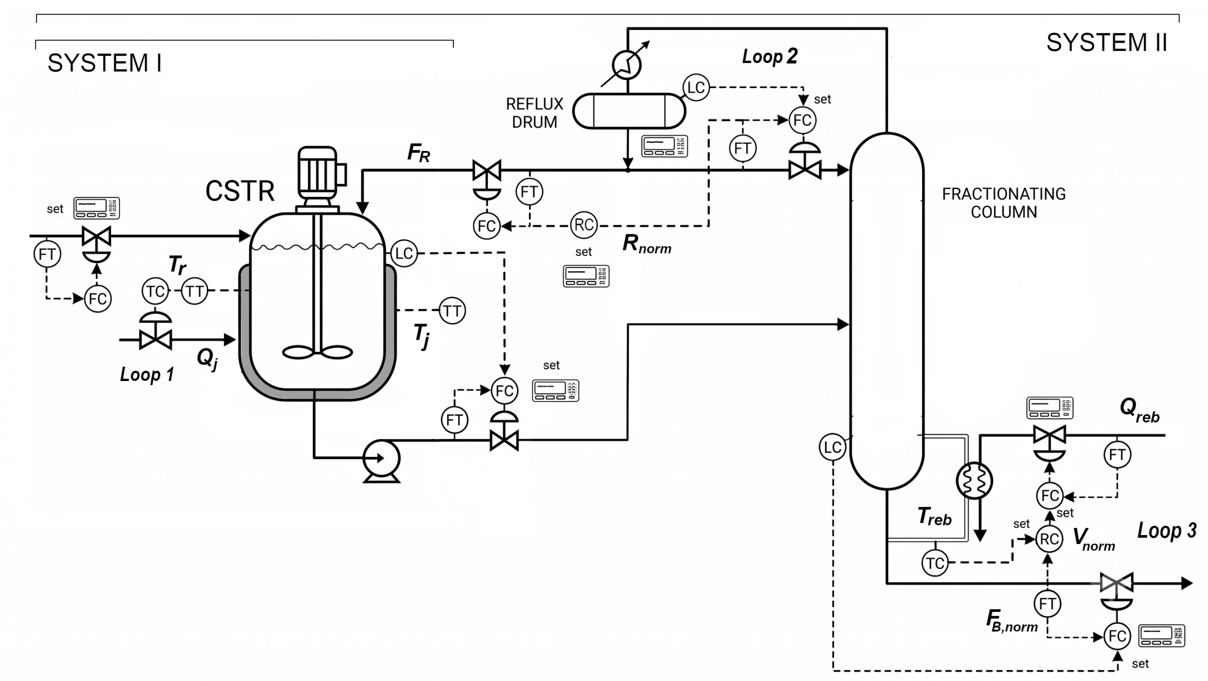

FIG. 1: Schematic of the two process systems studied. System I (left) is a single propylene oxide CSTR under PI temperature control. System II (full diagram) is the Luyben reactor-column-recycle benchmark [1] consisting of a CSTR coupled to a distillation column with liquid recycle and three decentralized PI loops.

A fixed operating setpoint with a high-gain controller implies near-zero spectral excitation of the controlled variable in the frequency band where fault-parameter information is concentrated, driving the local Fisher information for thermally-masked parameters toward zero. The practical consequence for fault diagnosis is twofold: some parameters are fundamentally harder to estimate than others under closed-loop operation, and apparent joint non-identifiabilities that emerge in compressed feature representations may be artefacts of lossy temporal aggregation rather than properties of the plant [23]. Characterising and distinguishing these two effects (scalar information reduction versus representation artefact) is the central methodological problem this work addresses. Simulation-based inference (SBI) [24] is an attractive inference method in this setting because it can approximate full Bayesian posteriors from simulator-generated data without requiring a tractable likelihood. Its amortised form is especially relevant for condition monitoring, where many observation windows must be processed repeatedly after a single offline training stage.

This paper makes four contributions: *1.* It shows that faults in a PI-controlled CSTR can be recovered accurately in non-saturating regimes, while jacket fouling remains harder to quantify under saturation and temperature-measurement bias. *2.* It demonstrates amortised SBI for plant-wide fault identification in a closed-loop reactor-column-recycle benchmark with five degradation parameters, recycle coupling, and three decentralised PI loops. *3.* It introduces a raw-trajectory Fisher information diagnostic for distinguishing representation artefacts from plant-level non-identifiability. *4.* It shows that the operational severity of a parameter confound depends on the downstream fault taxonomy, not only on the Fisher-information coupling magnitude.

II. SYSTEM I: CSTR WITH PI TEMPERATURE CONTROL

Two systems of increasing complexity are studied. As shown in Figure 1, System I can be considered a sub-system of System II, where System I is a single CSTR under proportional-integral temperature control, drawn from the canonical nonlinear CSTR benchmark of Fogler [25]. Specifically, we consider acid-catalysed liquid-phase hydrolysis of propylene oxide (PO) to propylene glycol. The reaction is pseudo-first-order in PO concentration, irreversible, and mildly exothermic following an Arrhenius rate law. Kinetic parameters are taken from Fogler (2016), Module 13 [25], whose values are traceable to the original experimental measurements of Furusawa *et al.* [26] for the ring-opening of propylene oxide in dilute aqueous acid solution.

A. Open-Loop Model

In the absence of feedback control, the reactor is characterised by three state variables: reactant concentration

C [mol L^{-1}], reactor temperature T [K], and jacket temperature T_c [K]. With coolant flow rate Q_c treated as a fixed external input, the energy-balanced material model is

$$\frac{dC}{dt} = \frac{Q}{V}\big(C_i - C\big) - k(T)\,C, \tag{1a}$$

$$\frac{dT}{dt} = \frac{Q}{V}\big(T_i - T\big) - \frac{\Delta H_r}{\rho C_p}\,k(T)\,C - \frac{UA}{\rho C_p V}\big(T - T_c\big), \tag{1b}$$

$$\frac{dT_c}{dt} = \frac{Q_c}{V_c}\big(T_{ci} - T_c\big) + \frac{UA}{\rho_c C_{pc} V_c}\big(T - T_c\big). \tag{1c}$$

The reaction is exothermic, with $\Delta H_r = -84{,}666$ J mol^{-1}. Under the sign convention of Eq. (1b), the heat-release contribution is therefore positive [25, 27, 28]. The overall heat transfer coefficient-area product UA [cal min^{-1} K^{-1}] couples the reactor and jacket energy balances. All nominal parameter values are listed in Table I.

The open-loop steady state is found by setting all time derivatives to zero and solving the resulting nonlinear algebraic system. At the nominal operating conditions (Table I) and $Q_c = Q_{c0} = 80$ L min^{-1}, the system has a unique physically meaningful steady state at $C_{\rm ss} \approx 0.018$ mol L^{-1}, $T_{\rm ss} \approx 312.5$ K, $T_{c,\rm ss} \approx 299.1$ K, corresponding to a single-pass conversion of approximately 98%. The open-loop model captures the essential coupling between reactor exothermicity and jacket cooling, and serves as the reference for understanding how feedback control modifies the observable system behaviour.

B. Closed-Loop Model

The closed-loop model employed in this work augments the open-loop system in Eq.(1) with a proportional-integral (PI) controller that manipulates the coolant flow rate Q_c to regulate reactor temperature at a setpoint $T_{\rm sp}$. In addition, the anti-windup integrator I [K min] accumulates the temperature error whenever the actuator is not saturated:

$$\frac{dI}{dt} = \big(T - T_{\rm sp}\big)\cdot \mathbf{1}\{Q_{c,\min} < Q_c < Q_{c,\max}\}\,, \tag{2}$$

providing conditional anti-windup. The integrator accumulation is suspended whenever the coolant valve is fully closed or fully open thereby preventing integrator windup during periods of actuator saturation [30].

The coolant flow rate Q_c [L min^{-1}] is computed from the parallel-form PI expression with actuator limits:

$$Q_c = \mathrm{clip}\bigg(Q_{c0} + K_p\big(T - T_{\rm sp}\big) + \frac{I}{\tau_i},\ Q_{c,\min},\ Q_{c,\max}\bigg)\,, \tag{3}$$

where $K_p = 150$ (L min^{-1}) K^{-1} is the proportional gain, $\tau_i = 10$ min is the integral time constant, $Q_{c0} = 80$ L min^{-1} is the bias (the nominal coolant flow at zero error), and $[Q_{c,\min}, Q_{c,\max}] = [0, 400]$ L min^{-1} defines the physical actuator range. A positive gain is used throughout since an increase in temperature above setpoint demands an increase in coolant flow (direct-acting on the energy balance). Under non-saturating, steady-state PI control (i.e. $Q_{c,\min} < Q_c < Q_{c,\max}$, $dI/dt = 0$), integral action drives the offset to zero and consequently, the closed-loop steady state satisfies $T_{\rm ss} = T_{\rm sp}$ exactly, independently of the values of the fault parameters. This is the fundamental closed-loop masking mechanism which we will analyse quantitatively in Section V B, where the PI controller actively suppresses the temperature signature of any parametric degradation that the energy balance would otherwise reveal.

At nominal conditions, the healthy closed-loop steady state satisfies $C_{\rm ss} \approx 0.018$ mol L^{-1}, $T_{\rm ss} = 312.5$ K (enforced by integral action), $T_{c,\rm ss} \approx 299.1$ K, and $Q_{c,\rm ss} \approx 48$ L min^{-1}. The primary observable consequence of closed-loop operation is that changes in the heat transfer or reaction parameters shift the steady-state coolant flow $Q_{c,\rm ss}$, not the reactor temperature. Jacket fouling (reduced UA) requires higher coolant flow to maintain the same temperature, while catalyst deactivation (reduced reaction rate) generates less heat and therefore requires lower coolant flow. These qualitatively distinct signatures in the actuator output Q_c are the primary diagnostic signal exploited by the inference framework developed in Section IV.

Two independent multiplicative degradation factors are introduced to represent the two dominant failure modes. The catalyst activity factor $\alpha \in [0.4, 1.2]$ scales the effective reaction rate constant, $k_{\rm eff}(T) = \alpha\, k_0 \exp(-E_a/RT)$ where $\alpha = 1$ represents a fresh catalyst, while $\alpha < 1$ corresponds to partial deactivation through fouling, sintering, or poisoning. The jacket fouling factor $\beta \in [0.4, 1.2]$ scales the effective heat transfer coefficient-area product, $(UA)_{\rm eff} = \beta \cdot UA$; $\beta = 1$ corresponds to a clean jacket, while $\beta < 1$ models progressive fouling of the heat transfer surface through scale formation, biofouling, or coking deposits.

Incorporating these factors, the degraded closed-loop model replaces the concentration and temperature state equations with

$$\frac{dC}{dt} = \frac{Q}{V}\big(C_i - C\big) - \alpha\, k\, C, \tag{4a}$$

$$\frac{dT}{dt} = \frac{Q}{V}\big(T_i - T\big) - \frac{\Delta H_r}{\rho C_p}\,\alpha\, k(T)\,C - \frac{\beta\cdot UA}{\rho C_p V}\big(T - T_c\big), \tag{4b}$$

$$\frac{dT_c}{dt} = \frac{Q_c}{V_c}\big(T_{ci} - T_c\big) + \frac{\beta\cdot UA}{\rho_c C_{pc} V_c}\big(T - T_c\big), \tag{4c}$$

while the PI control law (3) and anti-windup integrator (2) remain unchanged. Note that β and UA appear exclusively as the product $\beta \cdot UA$ in the model equations (4). Consequently, they cannot be estimated

TABLE I: Nominal parameter values for System I (propylene oxide CSTR). Kinetic parameters from Fogler [25] (Module 13) based on Furusawa *et al.* [26], physical properties from Green and Southard [29], controller tuning from the present work.

Parameter	Symbol	Value	Units	Source
Reaction kinetics				
Pre-exponential factor	k_0	1.696×10^{13}	min^{-1}	Fogler [25]
Activation energy	E_a	75,362	$\mathrm{J\ mol}^{-1}$	Fogler [25]
Standard heat of reaction	ΔH_r	−84,666	$\mathrm{J\ mol}^{-1}$	Fogler [25], Nie *et al.* [27]
Universal gas constant	R	8.314	$\mathrm{J\ (mol\ K)}^{-1}$	—
Reactor geometry and feed conditions				
Reactor volume	V	500	L	This work
Jacket volume	V_c	40	L	This work
Volumetric feed flow	Q	40	$\mathrm{L\ min}^{-1}$	This work
Residence time	$\tau = V/Q$	12.5	min	Derived
Inlet PO concentration	C_i	0.97	$\mathrm{mol\ L}^{-1}$	This work
Feed temperature	T_i	297	K	This work
Coolant inlet temperature	T_{ci}	297	K	This work
Physical properties				
Reactor fluid density	ρ	1,000	$\mathrm{g\ L}^{-1}$	Green and Southard [29]
Reactor fluid heat capacity	C_p	1.0	$\mathrm{cal\ (g\ K)}^{-1}$	Green and Southard [29]
Coolant density	ρ_c	1,000	$\mathrm{g\ L}^{-1}$	Green and Southard [29]
Coolant heat capacity	C_{pc}	1.0	$\mathrm{cal\ (g\ K)}^{-1}$	Green and Southard [29]
Heat transfer coefficient-area	UA	12,500	$\mathrm{cal\ (min\ K)}^{-1}$	This work
PI controller				
Temperature setpoint	T_{sp}	312.5	K	This work
Proportional gain	K_p	150	$(\mathrm{L\ min}^{-1})\ \mathrm{K}^{-1}$	This work
Integral time constant	τ_i	10	min	This work
Coolant flow bias	Q_{c0}	80	$\mathrm{L\ min}^{-1}$	This work
Coolant flow minimum	$Q_{c,\mathrm{min}}$	0	$\mathrm{L\ min}^{-1}$	Physical
Coolant flow maximum	$Q_{c,\mathrm{max}}$	400	$\mathrm{L\ min}^{-1}$	Physical
Fault parameter priors				
Catalyst activity	α	$\mathcal{U}(0.4, 1.2)$	—	This work
Jacket conductance	β	$\mathcal{U}(0.4, 1.2)$	—	This work

jointly from process observations alone. This is an instance of structural non-identifiability in the sense of Ljung [22]: no amount of data can resolve the two parameters independently without additional prior information. In this study, we consider the clean-service value $UA = 12{,}500\ \mathrm{cal\ (min\ K)}^{-1}$ a known design constant, and the inference target is the relative degradation factor β.

C. Fault Scenarios and Stochastic Dataset

We study eight operational scenarios spanning open- and closed-loop configurations and a range of fault severities. Their ground-truth parameters (α^*, β^*), operating modes, and experimental purpose are listed in Table II. Each scenario uses $N_r = 50$ independent stochastic replicates (60-minute windows, 120 time steps, four channels $[C, T, T_c, Q_c]$) for a total of 400 observation windows. All stochastic trajectories are generated via Euler–Maruyama integration with additive process and sensor noise starting from the nominal closed-loop steady state.

Scenario Sc 5 ($\beta^* = 0.4$) is of particular importance. At this fouling level the controller saturates ($Q_c = Q_{c,\mathrm{max}}$) for approximately 33% of each window, placing the system in a qualitatively different identifiability regime in which the actuator output carries no marginal information about further deterioration of β. Scenario Sc 7 introduces a fixed +2 K bias in the reactor-temperature measurement to test whether the inference framework can distinguish a biased temperature channel from mild jacket fouling. The bias changes the controller response in the same channels used to infer heat-transfer

TABLE II: Fault scenario definitions for System I. All closed-loop scenarios use the PI controller (Eq. (3)), open-loop scenarios fix $Q_c = Q_{c0} = 80$ L min^{-1}. $N_r = 50$ independent stochastic replicates per scenario, each comprising a 60-minute window at 0.5-min sample interval (120 time steps, 4 channels: C, T, T_c, Q_c).

ID	Name	Mode	Class	α^*	β^*	Bias	Primary purpose
Sc 0	Open-loop healthy	Open-loop	Healthy	1.00	1.00	—	Steady-state verification
Sc 1	Closed-loop healthy	Closed-loop	Healthy	1.00	1.00	—	SBI training / baseline
Sc 2	Jacket fouling	Closed-loop	Fouling	1.00	0.70	—	Primary inference: β
Sc 3	Catalyst decay	Closed-loop	Decay	0.70	1.00	—	Primary inference: α
Sc 4	Combined moderate	Closed-loop	Combined	0.85	0.85	—	Joint 2-D recovery
Sc 5	Severe fouling	Closed-loop	Fouling	1.00	0.40	—	Actuator saturation
Sc 6	Open-loop fault	Open-loop	Fouling	1.00	0.70	—	Mode-mismatch baseline
Sc 7	Fouling + temp. bias	Closed-loop	Fouling	1.00	0.85	+2 K	Measurement bias vs. fault disambiguation

loss.

The closed-loop scenarios (Sc 1–5, Sc 7) form the primary evaluation dataset while the two open-loop scenarios (Sc 0, Sc 6) serve as a steady-state verification check. Further details on all scenarios, sample observation trajectories, and the data-generation pipeline are provided in Section S1 of the Supporting Information.

III. SYSTEM II: LUYBEN REACTOR-COLUMN-RECYCLE BENCHMARK

The second system is the canonical reactor-separator-recycle benchmark of Wu *et al.* [1], Larsson *et al.* [31], Gyasi *et al.* [32], Seki and Naka [33]. It consists of a CSTR performing the irreversible liquid-phase reaction $A \to B$ coupled to a 20-tray distillation column whose A-rich distillate is recycled back to the reactor feed. The plant represents a standard industrially-motivated benchmark [34–36] for plant-wide control design. It exhibits the two diagnostic challenges of i) closed-loop temperature masking (Section V B) and ii) recycle-induced coupling between fault parameters. Three decentralised PI loops regulate the plant, and five degradation parameters must be estimated from process data. The reactor-jacket thermal dynamics and reactor heat-transfer degradation term used below are an extension introduced in this work on top of the Wu et al. topology and steady-state operating point; the published benchmark itself specifies the composition and holdup balances, but not a reactor energy balance.

A. Reactor-Column-Recycle Model

The reaction $A \to B$ follows first-order irreversible Arrhenius kinetics $k(T_r) = k_0 \exp(-E_a/RT_r)$ with nominal rate constant $k_{ss} = 0.33$ h^{-1} at the operating temperature $T_{sp} = 342.26$ K [1]. The reactor state vector $[z_A, T_r, T_j]$ evolves as

$$\frac{dz_A}{dt} = \frac{F_{in}}{M_r}\big(z_{A,in} - z_A\big) - k(T_r)\, z_A, \tag{5a}$$

$$\frac{dT_r}{dt} = \frac{F_{in}}{M_r}\big(T_{in} - T_r\big) + \frac{-\Delta H_r}{\rho C_p}\, k(T_r)\, z_A - \frac{UA_r}{\rho C_p V_r}\big(T_r - T_j\big), \tag{5b}$$

$$\frac{dT_j}{dt} = \frac{UA_r(T_r - T_j) - Q_j}{M_j\, C_{pj}}, \tag{5c}$$

where $F_{in} = F_0 + F_R$ is the total feed (fresh feed plus recycle), Q_j [Btu/h] is the actual coolant-side heat-removal duty delivered by Loop 1, and $M_r = 2{,}400$ lbmol is the reactor holdup. Nominal composition, flow, and operating-point values are taken from Wu *et al.* [1, Table 1] and are collected in Table III; the explicit reactor-jacket thermal layer is the extension described above. Section III C introduces five multiplicative degradation factors that modify Eqs. (5)–(7) to represent catalyst, heat-transfer, column, and feed faults.

The distillation column separates the reactor effluent into an A-rich distillate (recycle) and a B-rich bottoms product. The column hydraulic time constant ($\tau_{hyd} \approx 4$ s) is approximately 4,700× smaller than the reactor residence time ($\tau_r \approx 5.2$ h), so the column is modelled as a quasi-steady-state algebraic block using the Fenske–Underwood–Gilliland shortcut equations [37]. At clean-service separation efficiency, the effective relative volatility used by the shortcut equations equals the ideal value,

$$\alpha_{eff} = \alpha_{rel} = 2.0, \tag{6}$$

for which the column achieves $x_D = 0.95$ and $x_B = 0.011$ [1].

The recycle flow follows from the overall material balance:

$$F_R = D = d_{frac} F_{in}, \qquad z_{A,in} = \frac{F_0\, z_0 + F_R\, x_D}{F_0 + F_R}, \tag{7}$$

where D is the distillate flow, d_{frac} is the distillate fraction of the column feed, $F_0 = 460$ lbmol h^{-1} is the fresh

TABLE III: Nominal parameter values for System II (Wu 2003 reactor-column-recycle plant). Composition, flow, and operating-point values are sourced from Wu *et al.* [1, Table 1]; reactor-jacket thermal constants belong to the thermal extension introduced in this work.

Parameter	Symbol	Value	Units	Source
Reaction kinetics				
Rate constant at T_{sp}	k_{ss}	0.33	h^{-1}	Wu *et al.* 1
Activation energy	E_a	71.74	kJ mol^{-1}	Wu *et al.* 1
Heat of reaction	ΔH_r	-69.78	kJ mol^{-1}	Wu *et al.* 1
Reactor				
Operating temperature (setpoint)	T_{sp}	342.26	K	Wu *et al.* 1
Reactor holdup	M_r	2,400	lbmol	Wu *et al.* 1
Fresh feed rate	F_0	460	lbmol h^{-1}	Wu *et al.* 1
Fresh feed composition (A)	z_0	0.90	mol/mol	Wu *et al.* 1
Distillation column				
Number of trays	N_T	20	—	Wu *et al.* 1
Ideal relative volatility	α_{rel}	2.0	—	Wu *et al.* 1
Nominal reflux ratio	R	2.2	—	Wu *et al.* 1
Nominal distillate composition	x_D	0.95	mol A/mol	Wu *et al.* 1
Nominal bottoms composition	x_B	0.011	mol A/mol	Wu *et al.* 1
Liquid hydraulic time constant	τ_{hyd}	≈ 4	s	Derived

feed, and $z_0 = 0.90$ mol/mol is the fresh-feed A fraction [1]. Any reduction in per-pass conversion increases the A load on the column, raising the recycle flow and thereby reducing the reactor residence time leading to positive feedback known as the *snowball effect* [31, 35]. The normalised recycle flow $F_{R,\text{norm}} = F_R/F_{R,\text{nom}}$ is consequently the primary diagnostic indicator for faults that reduce reactor conversion, but it responds in the same direction to both catalyst decay and column separation loss, creating a natural parameter confound that the inference framework must resolve.

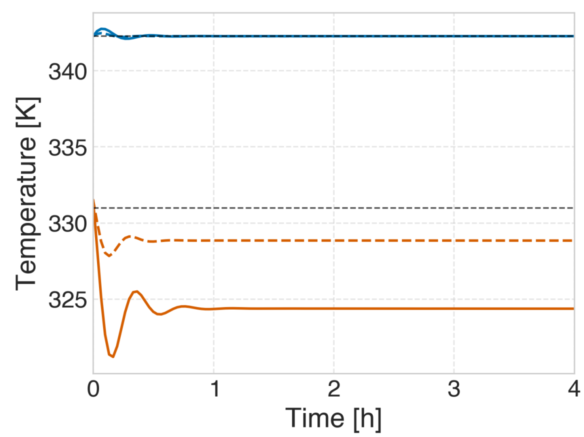

FIG. 2: Control masking in System II under jacket fouling (noiseless). W5: $\beta_r = 0.80$, dashed lines; W6: $\beta_r = 0.60$, full lines. Reactor temperature T_r is shown in blue, jacket temperature T_j in orange. Reactor temperature stays near the setpoint (dashed black line) under closed-loop operation regardless of fouling severity as Loop 1 masks the fault in the temperature channel. Jacket temperature T_j drops proportionally to the fouling magnitude, providing the primary secondary observable for β_r.

B. Control Structures

The control structure of System II is a three-loop decentralised PI controller, with no composition analysers [1, 31, 38, 39]. Loop 1 manipulates Q_j to hold $T_r = T_{\text{sp}}$. As in System I (Section V B), this integral action zeros the temperature sensitivity of any thermal fault parameter: $\partial T_{r,\text{ss}}/\partial\alpha \equiv \partial T_{r,\text{ss}}/\partial\beta_r \equiv 0$. The masking mechanism transfers exactly from the single CSTR to System II as confirmed by step tests that show T_r remain within $\approx$ 1–2 K of setpoint even under severe jacket fouling ($\beta_r = 0.60$), while the jacket temperature T_j provides the primary secondary diagnostic signal (Fig. 2). The other two loops regulate the distillation column: Loop 2 controls the reflux ratio R and Loop 3 controls the reboiler duty Q_{reb}. The affected channels are the F_R/F_0 flow ratio for Loop 2 and reboiler temperature control ($T_{\text{reb}} \to V$) for Loop 3. Nine channels are observed: $[T_r, T_j, Q_j, T_{\text{reb}}, Q_{\text{reb}}, F_{R,\text{norm}}, F_{B,\text{norm}}, R_{\text{norm}}, V_{\text{norm}}]$.

C. Fault Scenarios and Parameterisation

Five independent multiplicative degradation factors are inferred from process observations (Table IV). The two reactor-side parameters (α, β_r) from System I plus three parameters to capture column and feed degradation: $\eta_{\rm col}$, $\xi_{\rm reb}$, and $z_{A0,\rm eff}$. The first four parameters are multiplicative factors on the nominal physical constants, while $z_{A0,\rm eff}$ is an effective fresh-feed A fraction that accounts for variability in the feed composition or upstream separation efficiency. The lower bound $\eta_{\rm col} \geq 0.80$ is imposed by numerical stability of the quasi-steady-state column shortcut.

Incorporating all five degradation factors, the complete reactor–column–recycle model replaces Eqs. (5) and (7) with

$$\frac{dz_A}{dt} = \frac{F_{\rm in}}{M_r}\big(z_{A,\rm in} - z_A\big) - \alpha\, k(T_r)\, z_A, \tag{8a}$$

$$\frac{dT_r}{dt} = \frac{F_{\rm in}}{M_r}\big(T_{\rm in,mix} - T_r\big) + \frac{-\Delta H_r}{\rho C_p}\,\alpha\, k(T_r)\, z_A - \frac{\beta_r\, UA_r}{\rho C_p V_r}\big(T_r - T_j\big), \tag{8b}$$

$$\frac{dT_j}{dt} = \frac{\beta_r UA_r(T_r - T_j) - Q_j}{M_j\, C_{pj}}, \tag{8c}$$

where the fresh-feed composition fault propagates through the feed-mixing relation

$$z_{A,\rm in} = \frac{F_0\, z_{A0,\rm eff} + F_R\, x_D}{F_{\rm in}}, \qquad T_{\rm in,mix} = \frac{F_0\, T_{\rm in} + F_R\, T_r}{F_{\rm in}},$$

replacing the healthy-feed value $z_0 = 0.90$ of Eq. (7) by the inferred effective composition $z_{A0,\rm eff}$. The empirical column separation factor $\eta_{\rm col}$ scales the effective relative volatility away from its clean-service value, Eq. (6),

$$\alpha_{\rm eff} = 1 + \eta_{\rm col}\,(\alpha_{\rm rel} - 1), \tag{9}$$

which sets x_D, x_B, and hence the recycle flow F_R appearing in Eqs. (8a) and (III C). Finally, the reboiler conductance factor $\xi_{\rm reb}$ scales the heat duty that Loop 3 must supply to hold the column material balance at its quasi-steady state. The nominal column operating point ($x_{B,\rm nom}$, $F_{\rm in,nom}$, and the clean-service boilup reference) is taken from the Wu benchmark [1], while the following algebraic duty relation is an empirical quasi-steady proxy. Its form encodes four effects: nominal duty scales with throughput $F_{\rm in}/F_{\rm in,nom}$, with the boilup command $V_{\rm norm}$ from Loop 3, inversely with reboiler conductance $\xi_{\rm reb}$, and upward with bottoms impurity ($x_B - x_{B,\rm nom}$) and column-separation loss ($1 - \eta_{\rm col}$),

$$Q_{\rm reb} = \frac{Q_{\rm reb,nom}}{\xi_{\rm reb}}\frac{F_{\rm in}}{F_{\rm in,nom}} V_{\rm norm} \times \left[1 + 2\,(x_B - x_{B,\rm nom}) + 0.5\,(1 - \eta_{\rm col})\right], \tag{10}$$

where $F_{\rm in,nom} = F_0 + F_{R,\rm nom}$, $x_{B,\rm nom} = 0.011$, and $V_{\rm norm}$ is the normalised boilup effort commanded by Loop 3. A fouled reboiler ($\xi_{\rm reb} < 1$) therefore requires a proportionally larger duty $Q_{\rm reb}$ for the same throughput and boilup effort which is directly analogous to the role of β_r in the reactor jacket balance, Eq. (8c).

Table V lists the fourteen closed-loop fault scenarios studied through 30 stochastic replicates each giving rise to 420 observation windows. The scenarios cover individual single-parameter faults, compound multi-parameter faults, and a near-snowball-tipping-point scenario. Data generation follows the same Euler–Maruyama protocol as System I, adapted to 2-hour windows at 1-min sampling resolution (120 time steps). The process-noise levels and training hyperparameters are detailed in Section S7 of the Supporting Information.

TABLE IV: Fault parameters for System II (Wu 2003 recycle plant). All parameters equal 1.0 at clean-service nominal conditions. The physical design constants UA_r and UA_{reb} are treated as known and only the relative degradation factors are inferred.

Symbol	Physical meaning	Prior	Primary observable
α	Catalyst activity factor	$\mathcal{U}(0.4, 1.2)$	$F_{R,\mathrm{norm}}$, Q_j
β_r	Reactor jacket conductance factor	$\mathcal{U}(0.4, 1.2)$	T_j, Q_j
η_{col}	Empirical column separation factor	$\mathcal{U}(0.80, 1.0)$	T_{reb}, Q_{reb}
ξ_{reb}	Reboiler conductance factor	$\mathcal{U}(0.4, 1.2)$	Q_{reb}
$z_{A0,\mathrm{eff}}$	Effective fresh-feed A fraction	$\mathcal{U}(0.70, 0.95)$	All channels via reactor SS

TABLE V: Fault scenario catalogue for System II. All scenarios use 30 independent stochastic replicates per control structure, each comprising a 2-hour observation window at 1-min sample interval (120 time steps). Parameters equal 1.0 unless otherwise listed.

ID	Name	Fault unit	α	β_r	η_{col}	ξ_{reb}	z_{A0}	Primary purpose
W1	Healthy nominal	Healthy	1.00	1.00	1.00	1.00	0.90	SBI training / baseline
W2	Cat. decay mild	Reactor	0.85	1.00	1.00	1.00	0.90	Single-param: α
W3	Cat. decay severe	Reactor	0.65	1.00	1.00	1.00	0.90	Snowball onset
W4	Cat. near tipping	Reactor	0.55	1.00	1.00	1.00	0.90	Near snowball limit
W5	Jacket fouling mild	Reactor	1.00	0.80	1.00	1.00	0.90	Single-param: β_r
W6	Jacket fouling severe	Reactor	1.00	0.60	1.00	1.00	0.90	Single-param: β_r
W7	Col. sep. loss	Column	1.00	1.00	0.80	1.00	0.90	Single-param: η_{col}
W9	Reboiler fouling	Column	1.00	1.00	1.00	0.70	0.90	Single-param: ξ_{reb}
W10	Feed purity drop	Feed	1.00	1.00	1.00	1.00	0.78	Single-param: z_{A0}
W11	Cat. + jacket	Reactor	0.80	0.80	1.00	1.00	0.90	Compound: reactor faults
W12	Cat. + col. sep.	Multi	0.75	1.00	0.80	1.00	0.90	Compound: cross-unit
W13	Cat. + feed	Multi	0.80	1.00	1.00	1.00	0.80	Compound: reactor + feed
W15	Cat. near tipping	Reactor	0.58	1.00	0.90	1.00	0.90	Near snowball tipping
W16	Full multi-fault	Multi	0.75	0.80	0.80	0.85	0.90	All-unit degradation

IV. INFERENCE METHODOLOGY

The inference framework applied in this work is known as *simulation-based inference* (SBI), a likelihood-free inference methodology used when the likelihood function $p(\mathbf{y} \mid \boldsymbol{\theta})$ is analytically intractable or unavailable, but simulation of synthetic data $\mathbf{y}_{\mathrm{sim}} \sim p(\mathbf{y} \mid \boldsymbol{\theta})$ is feasible [24]. This methodology is widely used in scientific applications where the underlying physical model is known, but the complexity of the system makes it difficult to derive a closed-form expression for the likelihood In such cases, SBI allows for the estimation of posterior distributions over model parameters by leveraging simulations to generate synthetic datasets that can be compared to observed data. This is the case for the closed-loop fault-diagnosis problem considered here: the simulator includes nonlinear process dynamics, feedback control, stochastic disturbances, and sensor noise, while the likelihood of the resulting window-level summaries is not analytically tractable. The applied SBI framework comprises three successive stages: (i) A JAX-vectorised stochastic simulator that generates synthetic observation windows from the degraded closed-loop models. (ii) An amortised neural posterior estimator trained to approximate the Bayesian posterior $p(\boldsymbol{\theta} \mid \mathbf{s})$ over the degradation-parameter vector $\boldsymbol{\theta}$ given any realisation of the summary statistics. (iii) A threshold-and-unit-aggregation rule (Section IV A) that maps the joint posterior to a discrete fault class, reducing to four non-overlapping quadrants of the fault-parameter space for System I and to five unit-level classes for System II. The inference methodology is identical across System I (Section II) and System II (Section III) and differs only in the dimensionality of $\boldsymbol{\theta}$, the summary statistics $\mathbf{s}$, and the density-estimator training configuration (hyperparameters). The framework is implemented in the `sbi` library [40, 41]. Following the notation of Cranmer *et al.* [24], the estimator is operated in the *amortised* regime: the simulator is only queried at training time, and subsequent inference on any new 60-minute window requires only a single forward pass through the trained network, with typical latency below 20 ms.

A. Amortised Posterior Estimation and Classification

The inference target is an n-dimensional degradation-parameter vector $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$, with each component assigned an independent uniform prior distribution:

$$p(\boldsymbol{\theta}) = \bigotimes_{j=1}^{n} \mathcal{U}\big([\theta_{j,\mathrm{lo}},\, \theta_{j,\mathrm{hi}}]\big). \tag{11}$$

For System I, $n = 2$ and $\boldsymbol{\theta} = (\alpha, \beta)$, both assigned $\mathcal{U}(0.4, 1.2)$ (Table I). For System II, $n = 5$ and $\boldsymbol{\theta} = (\alpha, \beta_r, \eta_{\mathrm{col}}, \xi_{\mathrm{reb}}, z_{A0,\mathrm{eff}})$, with per-parameter bounds given in Table IV. For the parameters shared across both systems which are centred on a healthy value of 1 (α; β or β_r; ξ_{reb}), the upper bound of 1.2 serves dual purpose: it allows for a small positive bias in the estimated parameters, and it provides a numerical buffer around the healthy operating point, preventing the density estimator from being trained against a hard boundary at the most frequently sampled condition. The remaining two System II parameters have bounds set by other constraints (Table IV): η_{col}'s lower bound of 0.80 is set by the numerical stability of the quasi-steady-state column shortcut (Section III C), and $z_{A0,\mathrm{eff}}$'s bounds are centred on its own healthy value of 0.90, not 1.

A conditional density estimator $q_\phi(\boldsymbol{\theta} \mid \mathbf{s})$ is trained by minimising the expected forward Kullback–Leibler divergence over simulated $(\boldsymbol{\theta}, \mathbf{s})$ pairs drawn from the joint prior-predictive distribution,

$$\mathcal{L}(\phi) \;=\; -\,\mathbb{E}_{p(\boldsymbol{\theta})\, p(\mathbf{y}\mid\boldsymbol{\theta})}\Big[\log q_\phi\big(\boldsymbol{\theta} \mid \mathbf{s}(\mathbf{y})\big)\Big]. \tag{12}$$

Here, $\mathbf{s}(\mathbf{y})$ is the summary-statistic vector extracted from the raw observation window $\mathbf{y}$, and $p(\mathbf{y} \mid \boldsymbol{\theta})$ is the stochastic simulator (Section II C). The expectation is approximated by Monte Carlo sampling of $(\boldsymbol{\theta}, \mathbf{y})$ pairs from the prior and simulator, and the loss is minimised with respect to the density- estimator parameters ϕ using stochastic gradient descent. The trained estimator q_ϕ thus approximates $p(\boldsymbol{\theta} \mid \mathbf{s})$ for summary statistics reachable under the prior predictive distribution, enabling deployment on new observation windows without retraining. This is the single-round amortised formulation of sequential neural posterior estimation [SNPE-C; 42], in which the proposal distribution coincides with the prior so that no importance correction is required.

Following Forssell and Ljung [21], SNPE-C operates on the measured signal histories $\mathbf{x}(t)$ without requiring an explicit model of the feedback controller, making it a probabilistic analogue of the *direct method* of closed-loop system identification. Like the classical direct approach, it is universally applicable to nonlinear and saturating feedback and exploits the persistent in-loop process noise to partially recover parameter information that is structurally reduced by integral action [22].

The conditional distribution $q_\phi(\theta|s)$ is parameterised by a neural spline flow [43]. Network architectures, training budgets ($n_{sim} = 10,000$ for System I; $15,000$ for System II), and ensembling strategies were selected to obtain stable posterior estimates and acceptable calibration for the downstream diagnostic task. Full details of the network architectures, hyperparameter sweeps, and seed-stability strategies are provided in the Supporting Information (Sections S3 and S7.3). The Supporting Information also gives the exact simulation, summary-statistic, and calibration protocols used to reproduce the reported tables and figures.

Posterior calibration is assessed using simulation-based calibration [44], which tests whether the rank of the true parameter among posterior draws is uniformly distributed across independent test cases drawn from the prior. The full protocol, acceptance criteria, and rank histograms for each system's production posterior are given in the Supporting Information (Section S6, System I; Section S9.2, System II).

The joint posterior $q_\phi(\boldsymbol{\theta} \mid \mathbf{s})$ is mapped to a discrete fault class by a two-level rule applied to both systems, though the per-parameter threshold value now differs between them. First, each degradation parameter θ_j is compared against a relative threshold θ_{thr} and flagged as degraded if $\theta_j < \theta_{\mathrm{thr}}$. For System II, $\theta_{\mathrm{thr}} = 0.85$ (a 15% reduction relative to clean-service baseline) for the reactor- and column-side parameters (α, β_r, η_{col}, ξ_{reb}), exceeding typical run-to-run variability and providing a meaningful trigger for maintenance decisions; $z_{A0,\mathrm{eff}}$ is flagged relative to its own healthy value of 0.90, since a feed-purity fault is one-sided (always a reduction, never an increase), flagging a compositional deviation rather than a rate-constant or heat-transfer-coefficient reduction. For System I, $\theta_{\mathrm{thr}} = 0.90$ for both α and β. This threshold defines the operational fault-class decision rule and does not affect the posterior parameter estimates. Scenarios close to the threshold should therefore be interpreted as tests of both posterior uncertainty and the chosen decision rule.

Second, parameters are grouped into physical units such that for System I, each of the two parameters is its own unit; while for System II, reactor = $\{\alpha, \beta_r\}$, column = $\{\eta_{\mathrm{col}}, \xi_{\mathrm{reb}}\}$, feed = $\{z_{A0,\mathrm{eff}}\}$. A unit is flagged degraded if any of its member parameters is degraded. The reported class is the combination of degraded units with the greatest posterior mass: *healthy* (no unit degraded), the corresponding unit label if exactly one unit is degraded, or *multi* if more than one unit is degraded simultaneously. The associated confidence is given by the posterior mass assigned to the reported class. For System I, with two parameters each forming its own unit, this reduces to four non-overlapping quadrants of the (α, β) plane: *healthy*, *fouling-dominant*, *decay-dominant*, and *multi*. For System II it produces five classes (healthy, reactor, column, feed, multi), detailed in Section VI B. For the feed parameter this posterior-decision rule corresponds to $z_{A0,\mathrm{eff}} < 0.85 \times 0.90 = 0.765$. This posterior-decision threshold is distinct from the scenario-design labels in Table V, which define the intended fault unit of each synthetic case before posterior classification is applied.

V. RESULTS: SYSTEM I — PROPYLENE OXIDE CSTR

A. Model Validation and Fault Classification

Each replicate observation window $\mathbf{x} \in \mathbb{R}^{120\times 4}$ is compressed to a 29-dimensional summary-statistic vector $\mathbf{s}(\mathbf{x}) \in \mathbb{R}^{29}$ before inference. Most entries are generic time-series and control-effort descriptors (Table S1), while two are physics-informed proxies:

$$s_{UA} = \frac{\bar{T} - \bar{T}_c}{\bar{Q}_c}, \tag{13}$$

$$s_{k_0} = \ln\left(\frac{\bar{C}}{C_i - \bar{C}}\right), \tag{14}$$

where $\bar{(\cdot)}$ denotes the whole-window mean. The feature s_{UA} is a conductance-sensitive descriptor: lower effective heat transfer generally requires a larger temperature difference per unit coolant effort. Analogously, the feature s_{k_0} is motivated by the component balance:

$$s_{k_0} = \ln(Q/V) - \ln\alpha - \ln k(T), \tag{15}$$

to be a simplified reaction-rate-sensitive proxy.

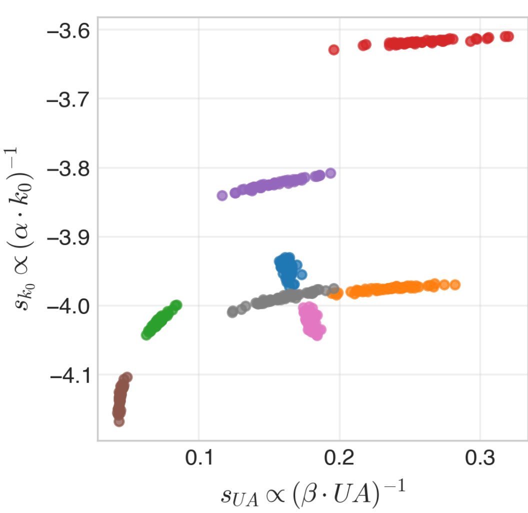

FIG. 3: Scenario separation in the two physics-informed summary features. Each point represents one of the 50 stochastic replicates with colour indicating scenario (Sc0-Sc7). The x-axis encodes $s_{UA} \propto (\beta \cdot UA)^{-1}$, the y-axis encodes $s_{k_0} \propto (\alpha \cdot k_0)^{-1}$. Closed loop scenarios (Sc1-Sc5) appear as thin strands while open-loop scenarios (Sc0, Sc6) are more clustered. Scatter within each cluster is due to process and sensor noise.

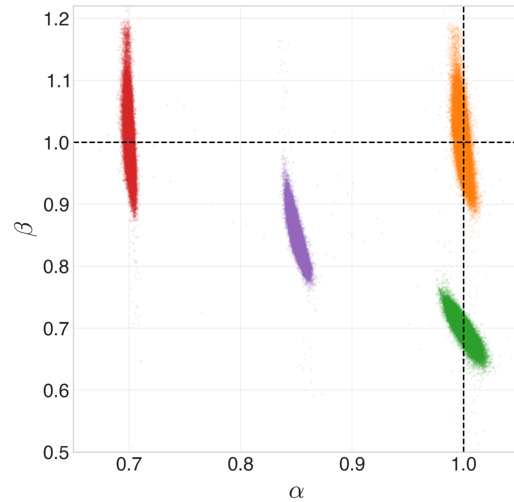

FIG. 4: Joint posterior samples for the primary System I closed loop scenarios: Sc1 (normal operation, orange), Sc2 (fouling, green), Sc3 (catalyst decay, red), Sc4 (combined, purple). Each panel pools posterior samples for one scenario and plots the inferred degradation parameters α and β. Dashed black lines mark the true scenario values. The narrow vertical structure of the posteriors indicates that α is generally tightly constrained, while β varies more strongly across the fault scenarios.

Simulator coverage of the evaluation regime was confirmed through prior predictive checks over 500 parameter vectors drawn from the prior, spanning the observed data range for all 29 summary features across every closed-loop scenario [24] (Fig. S1 of the Supporting Information). Posterior calibration with 500 test cases, drawn and evaluated under the matched (scenario-specific warm-start) protocol used throughout, does not reject rank uniformity at the 5% level (Kolmogorov-Smirnov (KS) p-values of 0.21 for α and 0.13 for β; two-sample classifier scores of 0.53 and 0.52, close to the uninformative baseline of 0.5). Rank histograms, SBC implementation details, and the matched-protocol calibration interpretation are given in Supporting Information Section S6. We evaluate fault separability in two complementary ways. First, as a diagnostic check on the summary representation, we train a supervised Linear Discriminant Analysis (LDA) classifier on the simulated scenario labels [45]. This tests whether the summaries contain enough information to separate the known fault classes. Second, for the SBI posterior itself, fault classes are assigned without historical labelled plant faults by applying the posterior threshold rule defined in Section IV A.

The two-feature $\{s_{UA}, s_{k_0}\}$ proxy set was found to be sufficient for near-perfect closed-loop LDA classification (99.3% accuracy; macro-F_1 = 0.993), as expected from Figure 3. The full 29-feature set removed the remain-

TABLE VI: System I per-scenario evaluation results for the production NSF posterior (50 replicates per scenario). $\hat{\alpha}$ and $\hat{\beta}$ are posterior means, 90% CI coverage gives the fraction of replicates for which the 90% credible interval contains the true value, F_1 is the per-scenario fault-classification F_1 score. Open-loop scenarios (Sc 0, Sc 6) are evaluated with a closed-loop-trained posterior and therefore reflect the mode-mismatch penalty discussed in Section V A. The +2 K temperature-measurement bias pulls $\hat{\beta}$ upward toward the 0.90 classification boundary; this is a genuine measurement-bias/fault confound.

ID	Name	α^*	β^*	$\hat{\alpha}$	$\hat{\beta}$	CI	F_1
Sc 0	Open-loop healthy	1.00	1.00	0.96	0.91	0.04	0.75[a]
Sc 1	Closed-loop healthy	1.00	1.00	1.00	1.00	0.90	1.00
Sc 2	Jacket fouling	1.00	0.70	1.00	0.70	0.86	1.00
Sc 3	Catalyst decay	0.70	1.00	0.70	1.01	0.90	1.00
Sc 4	Combined moderate	0.85	0.85	0.85	0.85	0.94	1.00
Sc 5	Severe fouling	1.00	0.40	1.02	0.44	0.62	1.00
Sc 6	Open-loop fault	1.00	0.70	1.02	0.93	0.00	0.04[a]
Sc 7	Fouling + sensor drift	1.00	0.85	0.97	0.89	0.52	0.67
Macro-F1 (closed-loop scenarios only)							**0.927**

[a] Evaluated with a closed-loop-trained posterior; open-loop mode mismatch is expected and intentional (see Section V A).

ing boundary errors, giving perfect LDA classification for both closed- and open-loop scenarios. Separately, mapping SBI posterior samples to the four threshold-defined fault regions gave a closed-loop macro-F_1 of 0.927 (Table VI). Coverage (90% of confidence interval) is close to its nominal level for the non-saturating matched closed-loop scenarios, but falls to 0.62 for Sc5. In this severe-fouling case, the coolant actuator is saturated for a substantial fraction of each window, reducing sensitivity to further changes in β and producing under-dispersed marginal intervals. The lower score for Sc 7 is due to a genuine confound between the temperature-measurement bias and the β degradation, which pulls the posterior mean upward toward the 0.90 classification threshold. This is a confound between the effect of the fault and the effect of the biased measurement channel.

The mode-mismatch experiment (Sc 2 evaluated with a posterior trained on open-loop data) quantifies the inferential cost of ignoring the feedback structure. The open-loop-trained posterior yields a Wasserstein-1 distance on β of 0.580, compared with 0.018 for the matched closed-loop posterior, a ∼33-fold increase; the CRPS on β increases by a similar ∼67-fold factor (0.579 vs. 0.009). The open-loop-trained posterior, never having seen a controller respond to a fault, has no mechanism to attribute the closed-loop Q_c rise to anything but severe fouling, collapsing its β estimate toward 0.15 (true value 0.70). When the situation is reversed (closed-loop-trained posterior applied to open-loop observations, Sc 6), fault classification accuracy collapses from 1.00 (the correctly-specified open-loop-trained posterior) to 0.04 as the closed-loop posterior assigns substantial mass above the $\beta = 0.90$ threshold because it has learned to interpret constant Q_c trajectories as evidence of healthy operation. Neither direction of mismatch is correctable within the single-round amortised framework where matched training is a necessary condition.

B. Local Sensitivity and Practical Identifiability

We assess local practical identifiability using the Gaussian sensitivity-information approximation

$$\widetilde{\mathbf{I}}(\boldsymbol{\theta}) = \mathbf{J}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\mathbf{J}, \tag{16}$$

where $\boldsymbol{\theta} = (\alpha, \beta)$,

$$\mathbf{J}_{ij} = \frac{\partial \mathbb{E}[s_i \mid \boldsymbol{\theta}]}{\partial \theta_j} \tag{17}$$

is estimated by finite differences, and $\boldsymbol{\Sigma}$ is the covariance matrix of the summary statistics estimated from stochastic simulator replicates at the same operating point. Equation (16) coincides with the Fisher information for a Gaussian summary model with parameter-independent covariance. Because the engineered summaries are correlated, nonlinear, and not guaranteed to be homoscedastic, we interpret $\widetilde{\mathbf{I}}$ as a local sensitivity and conditioning diagnostic, rather than as an exact Fisher information matrix or proof of global identifiability.

Using the full 29-dimensional summary representation and its estimated full covariance, the healthy operating point gives

$$\frac{\widetilde{I}_{\alpha\alpha}}{\widetilde{I}_{\beta\beta}} \approx 236. \tag{18}$$

At the Sc 2 jacket-fouling condition ($\alpha = 1$, $\beta = 0.70$), the same calculation gives a ratio of approximately 11.6. Across the tested range $\beta = 0.4$–1.0, the ratio varies from 5.4 to 168. The local sensitivity asymmetry is therefore strongly operating-point dependent rather than a fixed structural property. Near actuator saturation, these values should be interpreted cautiously because coolant-flow clipping makes the local derivative less regular.

The physical source of the asymmetry is the PI temperature loop (Loop 1 in Figure 1). In the non-saturating steady-state limit, integral action enforces $T_{\mathrm{ss}} = T_{\mathrm{sp}}$, and therefore

$$\frac{\partial T_{\mathrm{ss}}}{\partial \beta} = 0. \tag{19}$$

The steady-state component balance consequently makes the concentration measurement primarily sensitive to catalyst activity α. Jacket conductance β is observed indirectly through jacket temperature, coolant-flow effort, disturbance-driven dynamics, and saturation behaviour. Feedback therefore redistributes the observable β signature to secondary channels; it does not remove all information about β from the measured trajectory. The consequence is a loss of local conditioning, not a loss of identifiability in the strict structural sense. Under closed-loop operation, small changes in β can be harder to distinguish from stochastic variation and controller compensation than comparable changes in α. The achievable precision for β is consequently more sensitive to the operating point, summary representation, noise model, and actuator regime. In the non-saturating scenarios, this weaker conditioning does not prevent accurate posterior recovery as, for example, for Sc 2 (Table VI).

Under severe fouling, coolant-flow saturation reduces sensitivity to further changes in β and introduces a non-regular inference regime. For Sc 5 ($\alpha^* = 1$, $\beta^* = 0.40$), all 50 replicates remain correctly classified as fouling-dominant, while the mean estimate is $\hat{\beta} = 0.436$. Thus, fault detection remains robust because the posterior mass lies well inside the fouling region. The 90% marginal credible-interval coverage for β nevertheless falls to 0.62 in Sc 5, indicating substantial scenario-specific undercoverage. This should be interpreted as a limitation of severity quantification and uncertainty calibration under saturation, rather than as evidence that β becomes structurally unidentifiable.
Scenario Sc 7 shows instead a different practical consequence of the same reduced conditioning. In this case the jacket fault is mild ($\beta^* = 0.85$) and the observation includes a fixed +2 K bias in the reactor-temperature measurement. The controller responds to the biased temperature measurement, so part of the closed-loop actuator response that would otherwise identify heat-transfer loss can also be explained by the measurement bias. The posterior consequently shifts toward the healthy side of the $\beta = 0.90$ classification boundary ($\hat{\beta} = 0.89$), reducing the Sc 7 fault-class F_1 score to 0.67 and the 90% coverage for β to 0.52. This is not a strict non-identifiability of β in the clean measurement model; it is a practical confound introduced by an unmodelled sensor bias acting in the same closed-loop channels used to infer mild fouling.

We next compare the corrected summary-SBI posterior with three model-based baselines on Sc 2: No-U-Turn Sampler (NUTS) Markov chain Monte Carlo [46, 47], the extended Kalman filter (EKF) [48], and the unscented Kalman filter (UKF) [49]. All reported estimates use the same Sc 2 observation set with $\beta^* = 0.70$, subject to the method-specific conditioning and likelihood assumptions described in Section S8 of the Supporting Information. Summary-SBI targets an approximation to $p(\boldsymbol{\theta} \mid \mathbf{s}(\mathbf{y}))$, whereas NUTS targets the posterior defined by the implemented trajectory likelihood and the filters propagate Gaussian state-parameter approximations. The comparison is therefore an operational benchmark, not a claim that all four methods approximate an identical posterior object. The results do not show a common bias direction across methods. Summary-SBI is approximately unbiased on Sc 2, with $\hat{\beta} = 0.6977$, bias $= -0.0023$, and MAE $= 0.0124$. NUTS, aggregated over the 32/50 windows for which it converged within the 3-hour per-window compute budget, is likewise approximately unbiased ($\hat{\beta} = 0.700$, bias $= -0.0002$, MAE $= 0.011$), whereas the EKF and UKF underestimate β by approximately 0.093.

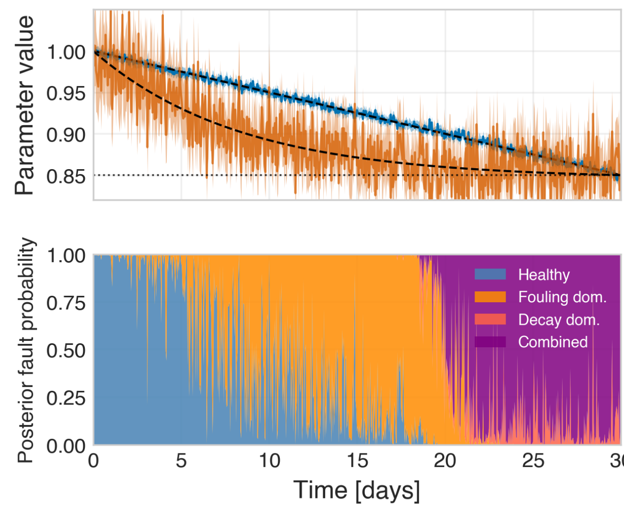

FIG. 5: 30-day sequential degradation tracking for System I (720 consecutive 60-min windows, α decay and β fouling, both reaching 0.85 at day 30). The upper panel overlays the amortised SBI posterior means and 90% credible intervals for $\alpha(t)$ and $\beta(t)$ on their true degradation profiles, with the classification threshold shown as a dotted reference line. The lower panel shows the corresponding posterior fault-class probabilities for each window, revealing the healthy-to-fouling and fouling-to-combined transitions as the stream crosses the decision thresholds.

These results separate two effects taking place. First, PI control suppresses the steady-state sensitivity of the controlled reactor temperature to jacket conductance and redistributes the remaining diagnostic information to the jacket-temperature and coolant-flow channels. This is a closed-loop practical-identifiability effect. Second, an amortised posterior trained on healthy-start fault-onset trajectories and evaluated on already-degraded steady-state trajectories can exhibit severe bias and miscalibration. This is a protocol-induced distribution-shift effect, not a property of the plant. Matching the simulator protocol to the intended monitoring regime removes most of the observed SBI bias while leaving the channel-sensitivity asymmetry as a separate phenomenon.

TABLE VII: Multi-method comparison on scenario Sc 2 (jacket fouling, $\alpha^* = 1.0$, $\beta^* = 0.70$). SBI, EKF, and UKF metrics are aggregated over all 50 stochastic observation windows. NUTS is also fit independently on all 50 windows under the same matched protocol, but under a fixed 3-hour per-window compute budget: 32/50 (64%) converged within budget and are used for the reported statistics; the remaining 18/50 (36%) did not complete within 3 hours and are excluded. This non-convergence rate is itself a finding, not merely a data-quality caveat: even after the protocol-artefact bias is corrected, NUTS's per-window computational reliability under the matched protocol is markedly worse than its already-documented raw speed disadvantage. ms/win for NUTS is the median wall time among the 32 converged windows; SBI/EKF/UKF's ms/win is a per-window mean, all measured under the compute environment described in Section S8.

Method	$\hat{\beta}$	Bias	MAE	σ	ms/win	Output
SBI	0.698	-0.002	0.012	0.015	24	Full posterior
NUTS ($n = 32/50$)	0.700	-0.0002	0.011	0.016	3.0×10^6	Full posterior
EKF	0.607	-0.093	0.097	0.077	32	Gaussian (μ, Σ)
UKF	0.607	-0.093	0.097	0.078	371	Gaussian (μ, Σ)

C. 30-Day Sequential Degradation Tracking

To demonstrate operational utility, the trained NSF posterior is applied to a synthetic 30-day monitoring stream consisting of 720 consecutive 60-minute observation windows. The degradation profile follows a linear catalyst decay $\alpha(t) = 1 - 0.15\,(t/720)$ and a Kern–Seaton [50] (asymptotic) fouling curve $\beta(t) = 1 - b\,(1 - e^{-t/200})$, with b chosen so that $\beta(720) = \alpha(720)$ exactly, so that by the end of the monitoring period $\alpha^* = \beta^* = 0.85$ — a milder target than an earlier version of this scenario, chosen to match the mild-fault severity used elsewhere in this manuscript's scenario design (e.g. Sc 4, Table VI). Because the asymptotic $\beta(t)$ falls faster than the linear $\alpha(t)$ at early times, β crosses the classification threshold first (around day 8.7) and α later (around day 20), so the stream passes through a genuine *healthy* $\to$ *fouling-dominant* $\to$ *combined* progression. Each window is processed independently with no temporal prior propagation performed, making the tracker stateless and directly deployable in an existing process historian without bespoke integration.

The SBI tracker achieves $\mathrm{MAE}_\alpha = 0.0027$ and $\mathrm{MAE}_\beta = 0.0206$ over the full 30-day run, completing all 720 windows in 8.5 s (wall clock, CPU inference) — an amortisation speedup of approximately 38,900× relative to an equivalent per-window NUTS estimate. For comparison, EKF and UKF achieve $\mathrm{MAE}_\alpha = 0.0232/0.0249$ and $\mathrm{MAE}_\beta = 0.0652/0.0903$ respectively, roughly an order of magnitude higher α error and 3–4× higher β error than SBI, consistent with the genuine closed-loop information-deficit evidence discussed in Section V B. Fault classification is correct in 84.4% of windows overall; this aggregate masks a non-monotonic pattern that is itself informative: accuracy is lowest ($\approx 80\%$) during the two phases where the stream crosses a genuine classification-threshold boundary, and highest ($\approx 94\%$) once both parameters are unambiguously on the fault side of their thresholds.

VI. RESULTS: SYSTEM II — LUYBEN REACTOR-COLUMN-RECYCLE PLANT

For System II, the central finding is that the 66-D summary representation introduces apparent joint parameter couplings that are not plant-level non- identifiabilities under a raw-trajectory Fisher-information check. In contrast, reactor-temperature masking by Loop 1 remains a genuine representation-independent information loss.

A. Physics-Informed Summary Statistics for System II

Similar to System I, each 2-hour observation window is compressed into a 66-dimensional summary-statistic vector prior to passing it to the neural density estimator. The complete feature definitions, per-parameter mutual-information rankings, and PCA-based separability analyses are detailed in the Supporting Information.

Among these features, two physics-informed proxies carry much of the identifying information. The System II analogue of s_{UA} for System I (Eq. (13)), is given by:

$$s_{UA,r} = \frac{T_r - T_j}{Q_j} \propto (\beta_r \cdot UA_r)^{-1} \,. \tag{20}$$

Under Loop 1 PI control, $T_r \approx T_{\mathrm{sp}}$ and the numerator encodes the jacket temperature drop driven by fouling, while the denominator normalises by the controller output. The normalised recycle flow deviation

$$s_{F_R} = F_{R,\mathrm{norm}} - 1 \tag{21}$$

provides the primary snowball indicator. Any reduction in per-pass conversion (lower α or η_{col}) drives $F_{R,\mathrm{norm}}$ above 1. Because both parameters raise s_{F_R} in the same direction, s_{F_R} alone cannot distinguish catalyst decay from column-efficiency loss and a confound is to be expected. The identifiability analysis in Section VI C investigates this directly.

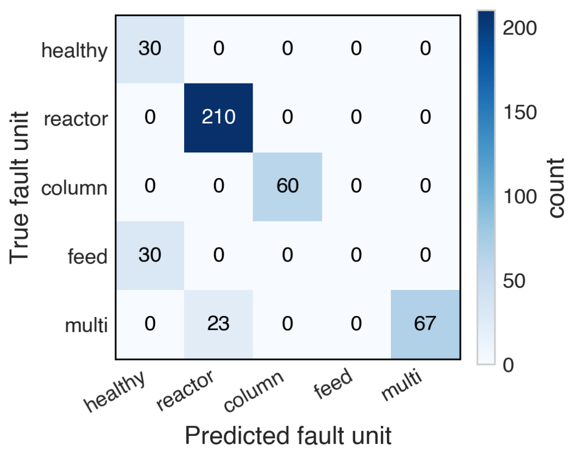

FIG. 6: Unit-level fault confusion matrix for System II (14 scenarios, 30 replicates each). Rows: true fault unit; columns: predicted fault unit. Reactor and column faults are reliably detected. Feed-fault scenarios (W10, W13) are predominantly misclassified as reactor or compound faults as a consequence of the representation-level $(\alpha, z_{A0,\text{eff}})$ confound identified in Section VI C 3.

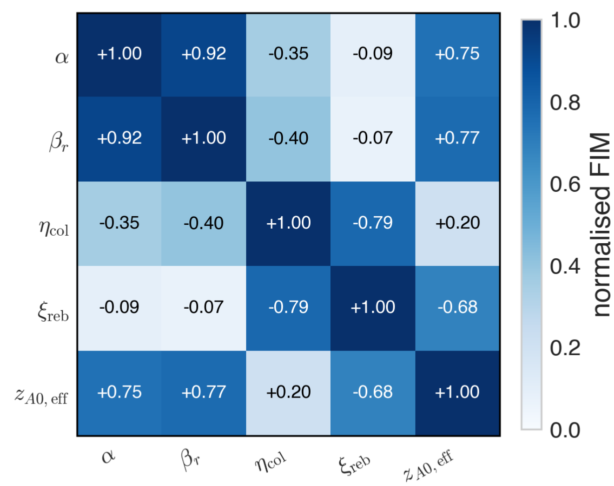

FIG. 7: Normalised 5×5 Fisher information matrix at System II nominal operating point. Entries show the signed, normalised parameter coupling at the nominal operating point, with colour intensity indicating absolute information coupling strength. The strong positive $\alpha - \beta_r$ off-diagonal term highlights a dominant identifiability confounding between reactor degradation and reactor heat-transfer loss, while the remaining structure shows weaker or more localized couplings among column efficiency, reboiler efficiency, and feed composition. The high off-diagonal +0.93 for (α, β_r) under the full feature set is investigated as a potential joint degeneracy in Sections VI C 1–VI C 3.

B. Fault Classification

The production NSF posterior was applied to the 14 closed-loop System II fault scenarios using the unit-level posterior-mass rule defined in Section IV A. Classification is reported at the fault-unit level because operational decisions concern whether the reactor, column, feed system, or multiple units are degraded, rather than only which individual parameter crossed the threshold.

Overall unit-level classification accuracy is 87.4%, with macro-F1 = 0.694. Performance is uneven across units: reactor and column faults are detected reliably, whereas feed faults fail under the present representation. Per-unit results are summarised in Table VIII and the confusion matrix is shown in Figure 6.

Breaking down the performance by functional unit reveals how representation artefacts impact downstream diagnosis. Reactor and column faults are detected reliably. Reactor scenarios are correctly classified despite parameter-level ambiguity between α and β_r, because both parameters map to the same reactor unit. Column scenarios are also classified reliably, including the cross-unit W12 case. In contrast, the feed-purity scenario W10 is misclassified in all 30 replicates, indicating that the present representation does not support reliable feed-fault attribution.

Table IX places the SBI classification alongside the baseline EKF estimates. The EKF structurally cannot observe $z_{A0,\text{eff}}$ (estimating only $[\alpha, \beta_r, \eta_{\text{col}}]$), offering no comparison point at W10. Where the EKF does produce estimates, two distinct regimes emerge. At the near-snowball scenarios W12 and W15 ($\alpha = 0.75$ and $\alpha = 0.58$ respectively), the EKF achieves only 3% empirical coverage on α,consistent with a failure of local Gaussian linearisation under strong non-linear recycle dynamics near the tipping point. SBI, conversely, achieves 100% coverage at W15. At W11, the EKF's 0% coverage is a tuning-sensitive pitfall rather than evidence of a genuine degeneracy, as an independently re-tuned EKF with raw-trajectory access recovers both parameters to $< 2\%$ error (Section VI C 2). Crucially, SBI's unit-level classification remains correct across all three of these scenarios (W11, W12, W15) regardless of baseline tuning sensitivities.

C. Structural Identifiability Analysis

System II raises a different identifiability question from System I. Loop 1 masks reactor-temperature sensitivity in the same way as the CSTR controller, but the dominant difficulty in System II is not only scalar information loss. The 66-D summary representation also introduces apparent joint couplings between degradation parameters.

To distinguish plant-level non-identifiability from representation-induced coupling, we recompute the local FIM for each candidate confound using raw, unaggregated trajectories of the already observed channels at

TABLE VIII: Unit-level fault classification results for System II (14 scenarios, 30 replicates for the production NSF posterior, $\theta_{\text{thr}} = 0.85$). Per-class F_1 is the harmonic mean of precision and recall across all replicates in each fault-unit category.

Fault unit	Scenarios	F_1	Note
Healthy	W1	0.667	Some healthy replicates border the threshold
Reactor	W2–W6, W11, W15	0.948	Near-perfect despite (α, β_r) corr = 0.998
Column	W7, W9	1.000	η_{col} and ξ_{reb} well-identified
Feed	W10	0.000	Artefact 3 confound with α (Section VI C 3)
Multi	W12, W13, W16	0.854	W13 weak (7/30): feed+reactor compound, Artefact 3
Overall macro-F1		0.694	87.4% accuracy

TABLE IX: SBI vs. EKF at the scenarios used to illustrate fault-classification robustness (System II). The augmented EKF (Section VI D) estimates only $[\alpha, \beta_r, \eta_{\text{col}}]$ whereas ξ_{reb} and $z_{A0,\text{eff}}$ are outside its augmented state and are therefore unobservable to it by construction, not merely difficult to estimate.

Scenario	True fault unit	SBI classification	EKF estimate	EKF 90% coverage	Note
W10	Feed ($z_{A0,\text{eff}} = 0.75$)	Misclassified, $F_1 = 0.000$	—	—	$z_{A0,\text{eff}}$ not in EKF's augmented state
W11	Reactor ($\alpha = \beta_r = 0.80$)	Correct, 30/30	$\hat{\alpha} = 0.907$, $\hat{\beta}_r = 0.986$	0%	Tuning-sensitive; see Artefact 2
W12	Multi ($\alpha = 0.75$)	Correct, 30/30	$\hat{\alpha} \approx 1.19$	3%	Snowball nonlinearity
W15	Reactor ($\alpha = 0.58$)	Correct, 30/30	$\hat{\alpha} = 0.990$	3%	Near snowball tipping point

the same operating point. We compare normalized off-diagonal entries, Cramér-Rao standard deviations, and FIM condition numbers across the summary and raw representations. A substantial reduction in normalized coupling accompanied by improved conditioning indicates that the compressed representation has discarded locally distinguishing information. The test is local and information-theoretic: it identifies available information in the measured signals, but does not guarantee recovery by a finite-capacity amortised estimator. The RT-FIM check is computationally inexpensive relative to posterior training. At the replicate count used here, one representation at one operating point requires $\mathcal{O}(10^2)$ simulator calls, compared with $\mathcal{O}(10^4)$ simulations for the amortised posterior training.

1. *Artefact 1: Restricted-channel coupling between α and η_{col}*

The apparent $\alpha - \eta_{col}$ confound appears only when the analysis is restricted to recycle-flow-derived features. Both parameters affect $F_{R,\text{norm}}$ through conversion and recycle coupling, so this restricted representation produces an elongated ridge. Once the already measured reboiler channels, T_{reb} and Q_{reb}, are included in the full feature set, the ambiguity is removed: posterior correlations remain below 0.15 and the η_{col} 90% credible interval width is approximately 0.01-0.02 across the tested scenarios. This is therefore a restricted-channel artefact, not a plant-level non-identifiability. Consistent with this interpretation, scenario W12 is classified correctly as a compound reactor-column fault in all 30 replicates.

2. *Artefact 2: Temporal-aggregation confound between α and β_r*

The strongest summary-level confound is the reactor-side pair $\alpha - \beta_r$. At W11, the 66-D summary representation gives a posterior correlation of $+0.998$ and a FIM-normalized off-diagonal of approximately $+0.85$ to $+0.90$, indicating an almost one-dimensional ridge. The raw-trajectory FIM indicates that this coupling is not intrinsic to the measured plant signals. Recomputing the FIM on the unaggregated trajectories of the already observed reactor-side channels collapses the normalized off-diagonal to approximately zero. The condition number falls from 3.45×10^3 under the 66-D summaries to 5.34 under the raw trajectory, showing that the improvement is not merely a rotation of the uncertainty ellipse but a genuine local gain in parameter distinguishability. Feature decomposition indicates that whole-window statistics of the cooling duty Q_j dominate the summary-level coupling. This is physically plausible: both lower catalyst activity and lower jacket conductance are compensated by Loop 1 through changes in Q_j.

Whole-window means and quantiles preserve the common level shift but discard transient shape information that distinguishes the two mechanisms. This confound is therefore a temporal-aggregation artefact. However, it is not automatically solved by any raw-data estimator: the retuned EKF with raw-model access resolves the pair, whereas the tested CNN-SBI posterior does not. The RT-FIM result should therefore be interpreted as local information-theoretic recoverability, not as a guarantee of amortised SBI recovery.

3. Artefact 3: Composition-confound between reactor faults and feed purity

The feed-purity parameter $z_{A0,\text{eff}}$ shows strong summary-level coupling with reactor-side degradation at degraded operating points. For example, at the W2 operating point the α-$z_{A0,\text{eff}}$ FIM off-diagonal is approximately -0.89 under the 66-D representation, compared with approximately -0.12 at nominal conditions. The raw-trajectory FIM removes this local coupling: the corresponding off-diagonal falls to approximately -0.07, and the condition number decreases from 4.36×10^3 to 5.48. The analogous β_r-$z_{A0,\text{eff}}$ coupling at the W5 operating point shows the same pattern, with the condition number decreasing from 41.4 to 5.24. Thus, the feed-reactor coupling is also representation-induced rather than a plant-level non-identifiability.

Unlike Artefact 2, this representation artefact is operationally severe. The confused parameters belong to different fault units, so the representation-level ambiguity corrupts unit-level classification rather than only parameter attribution within the same unit. This explains why W10 feed faults are misclassified in all 30 replicates and the feed class has F1 = 0.000.

Raw-trajectory CNN posterior as a practical recovery test. A CNN-embedding SBI posterior was trained on raw 120×9 trajectories to test whether an amortised neural estimator could exploit the information identified by RT-FIM. The results are mixed. Artefact 1 remains resolved, and the α-$z_{A0,\text{eff}}$ correlation associated with Artefact 3 collapses from approximately $+0.99$ to near zero. However, Artefact 2 remains unresolved: the α-β_r posterior correlation remains above $+0.99$ across tested seeds and operating points.

The CNN posterior also fails to improve feed-fault classification. Feed F1 remains 0.000, although the failure mechanism changes: under the hand-crafted posterior, feed faults are masked by the α-$z_{A0,\text{eff}}$ correlation, while under the CNN posterior they are dominated by a systematic downward bias in α. Consequently, the CNN result reinforces the distinction between local information availability and practical amortised recovery. RT-FIM can show that the raw signals contain distinguishing information, but estimator architecture, training budget, and bias determine whether that information is actually used. This failure does not invalidate the plant-wide SBI framework; rather, it identifies feed attribution as the unit for which the current measurements, representation, or decision rule are insufficient.

Together, the three cases show that summary-induced confounds should not be interpreted solely by their Fisher-information magnitude. Artefacts 2 and 3 have comparable summary-level coupling, but their classification consequences are opposite. The α-β_r ambiguity is mostly benign for unit-level diagnosis because both parameters belong to the reactor unit. The α-$z_{A0,\text{eff}}$ ambiguity is severe because it crosses the reactor-feed unit boundary. Thus, the operational severity of a representation artefact depends on the fault taxonomy used for decision-making, not only on local parameter coupling.

D. 30-Day Sequential Degradation Tracking

The calibrated posterior is applied to a synthetic 30-day monitoring stream of 360 consecutive 2-hour observation windows, each processed independently (no temporal state carried between windows). The degradation profile follows a linear catalyst decay reaching $\alpha \approx 0.65$ and a Kern–Seaton fouling curve reaching $\beta_r \approx 0.90$ by day 30, with η_{col}, ξ_{reb}, and $z_{A0,\text{eff}}$ held at their healthy values throughout. Results are shown in Figure 8.

In this sequential tracking experiment, the EKF outperforms the amortised SBI posterior. This result should be interpreted as a limitation of the present per-window summary representation rather than as a general limitation of SBI. Control experiments (Supporting Information §S11) rule out any advantage from accumulating information across the 360-window run, and instead trace the gap to the EKF's privileged access to the raw, unaggregated trajectory through the exact ODE model consistent with Artefact 2 (Section VI C 2). The CNN-embedding posterior (Supporting Information Section S10) narrows part of this gap (MAE_{β_r} improves to 0.139) but not all of it (MAE_α worsens to 0.067), both still an order of magnitude short of the EKF. SBI remains substantially faster per window (15 ms vs. 25.9 ms for the EKF).

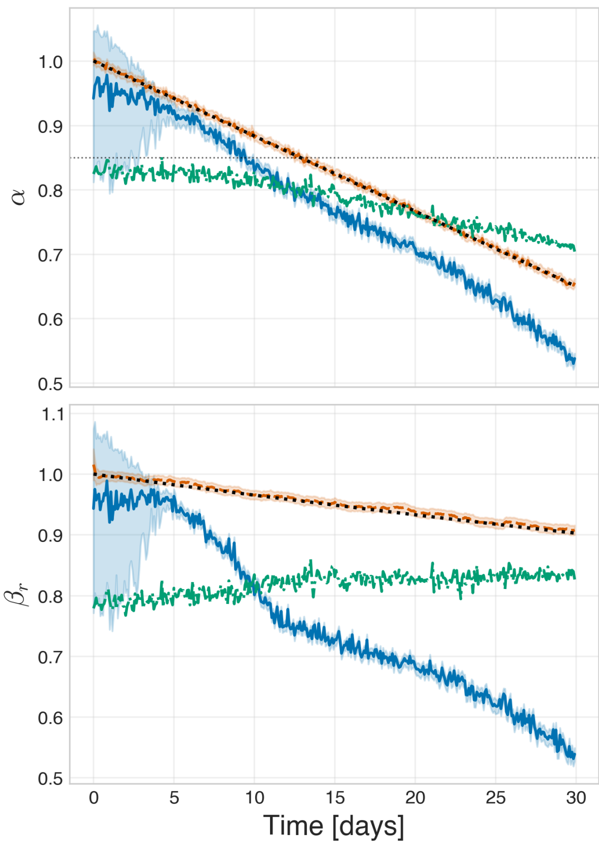

FIG. 8: Sequential System II degradation tracking over 30 days. The true trajectories for catalyst activity α and β_r and reactor heat-transfer factor are compared with independent amortised SBI estimates (blue), recursive EKF estimates (orange), and CNN-embedding SBI posterior means (green) across 360 two-hour windows. Shaded bands show 90% intervals for SBI and EKF only. The EKF tracks both slowly varying parameters closely, while the amortised SBI methods show larger bias and delayed response, especially for β_r, illustrating the difficulty of recovering gradual degradation from independent window-level posterior inference.

VII. DISCUSSION

A. One Genuine Limit, Three Representation Artefacts

The two systems separate two qualitatively different sources of diagnostic difficulty. The first is genuine closed-loop information loss: integral control removes the primary controlled-variable signature of thermal degradation. This effect appears strongly for β in System I and more mildly for α and β_r in System II. Because it is present in the raw trajectories and reproduced across inference methods, it cannot be removed by changing summary statistics.

The second source is representational. In System II, the 66-D summary representation creates apparent joint couplings that weaken or disappear under the RT-FIM check. These couplings are therefore not plant-level non-identifiabilities, although they may still harm practical SBI inference and fault classification. The distinction is operationally important: representation artefacts call for improved features, learned embeddings, or revised decision rules, whereas genuine closed-loop masking requires new measurements or deliberate excitation.
These results show that plant-wide SBI posteriors must be evaluated against the fault taxonomy used for operational decisions, not only against parameter-level identifiability metrics.

B. Why Amortised SBI Is the Appropriate Instrument for This Problem Class

SBI is useful for closed-loop plant-wide fault identification because the simulator is available but an explicit likelihood over multi-unit, feedback-controlled observation windows is not. Its main advantage over Gaussian filters is not universal accuracy, but the ability to represent joint, non-Gaussian posterior structure and to amortise inference across repeated monitoring windows. This is particularly valuable when the posterior geometry itself is diagnostically meaningful, as in the distinction between unit-benign and unit-severe confounds. However, the results also show that SBI is not uniformly superior to model-based estimators. In System II tracking, a tuned EKF with raw-model access outperforms the amortised summary-statistic posterior. Thus, the practical value of SBI is strongest when likelihood construction is difficult, posterior geometry is non-Gaussian, or per-window optimisation is undesirable, not when an exact model and well-tuned recursive estimator are available.

C. EKF Failure and Success Regimes

The EKF comparisons indicate that no universal ranking between SBI and classical estimators is appropriate. EKF failures near W12/W15 reflect the limits of local Gaussian linearisation under strong recycle nonlinearity, while the W11 failure is tuning-sensitive rather than structural. Conversely, in 30-day System II tracking, a well-tuned EKF with exact raw-model access outperforms the current per-window SBI posterior. Recursive pooling of biased per-window SBI posteriors does not solve this problem because it compounds systematic representation bias rather than averaging independent noise. The relevant distinction is therefore not SBI versus EKF in general, but amortised summary-based inference versus raw-trajectory model-based estimation under each operating regime.

VIII. SUMMARY AND CONCLUSIONS

This work demonstrates amortised SBI as a framework for Bayesian fault identification in closed-loop chemical processes, including a plant-wide reactor-column-recycle system with multiple interacting units and decentralized PI control.
In the PI-controlled CSTR, jacket fouling is intrinsically harder to estimate than catalyst activity because integral temperature control removes the primary temperature signature of heat-transfer degradation. Under the original training protocol, a β bias appeared in all of SBI, CNN-SBI, MCMC, EKF, and UKF; however, SBI, CNN-SBI, and MCMC shared a training/evaluation initial-condition mismatch (a protocol-induced inference artefact, not a plant property), and correcting it removes nearly all of their bias while EKF and UKF's bias — unaffected by the mismatch, since their filter state initialises from the observed data directly — remains. The persistent EKF/UKF bias is therefore the more credible remaining evidence of a genuine, though smaller than originally reported, closed-loop information deficit for β.

In the reactor-column-recycle benchmark, the 66-D summary representation produces three apparent joint confounds. Raw-trajectory FIM analysis shows that these are locally recoverable from the already observed signals and are therefore representation artefacts rather than plant-level non-identifiabilities. However, their downstream consequences differ: reactor-side ambiguity is benign for unit-level classification, whereas feed-reactor ambiguity collapses feed-fault detection.

The results show that SBI is useful not because it universally outperforms classical estimators, but because it provides joint posterior structure and amortised inference in settings where likelihood construction is difficult. Conversely, when an exact model and raw trajectory are available, a well-tuned EKF can outperform the current SBI implementation. Practically, representation artefacts call for richer features, learned embeddings, or additional decision rules, while genuine closed-loop masking requires new measurements or deliberate excitation. The present System II implementation therefore demonstrates plant-wide SBI fault identification for reactor and column faults, while identifying feed-fault attribution as the remaining limitation requiring richer measurements, less lossy representations, or revised decision rules.

IX. DATA AND CODE AVAILABILITY

The code and experiments used in this paper are available on GitHub at: `https://github.com/XXXXX`.

X. CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Simone Casolo: Writing, methodology, formal analysis, conceptualization. **Alexander J. Stasik**: Review & editing, supervision, methodology. **Signe Riemer-Sørensen**: Review & editing, supervision, methodology, conceptualization.

XI. DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

XII. ACKNOWLEDGEMENTS

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